Which Preference Gets Measured? Context and Channel Instability in Model Preference Audits

Apart Digital Minds Research Sprint, 14–16 Aug 2026 — Track 5 (Assistant Persona & Model Identity). Benji Berczi; implementation and analysis by Claude (Anthropic) under the author's direction (full disclosure in the technical report §6). Repository: github.com/benjibrcz/which-preference-gets-measured (all raw model outputs cached; analyses reproducible without API access).

Framing and contribution

The sprint asks whether frontier models possess genuine preferences or portray characters. We reframe it as a **measurement** question: **single-context, single-channel audits do not identify a context-independent model preference profile.** We began with a representation-versus-binding hypothesis, but a post-review non-agent control showed agenthood is not necessary; the robust result is broader — explicitly non-adopted, task-relevant context changes committed choices, while ownership, prediction, choice, and identity can disagree about the same configuration. A sequence of increasingly matched controls narrowed the mechanism and repeatedly corrected our own interpretation. Building directly on Gilg, Beckmann, Paleka & Butlin (arXiv:2605.13339) — who measured revealed task preferences and probed them internally, but did not test self-report, separate described from enacted personas, or measure dynamics — we measure the *same* 76 task-pair preferences through revealed choice, three verbal self-report formats, 0–10 task ratings, and an identity query, under a preregistered grid of persona framings, on four large instruction-tuned models from four labs (Gemma-3-27B, Llama-3.1-70B, gpt-4.1-mini, Qwen-2.5-72B). Beyond Gilg et al. (the battery/probe antecedent), this positions against Personascope (persona-adoption depth vs value drift), in-context persona-induction work, stated-vs-revealed LLM preference studies, and *Where Is the Mind?* (persona-based individuation) — our distinct move is the reverse dissociation plus the agenthood-agnostic control.

Preregistration and QC. The core grid and its analysis gates were preregistered before its data (PREREG.md); later extensions were sequentially preregistered, and the non-agent normative control was exploratory, added post-review. Factual invariant controls ≥ 0.98 everywhere; parse health ≥ 0.93 after a logged uniform parser extension; order counterbalanced; planted-signature calibration passed; headline estimates carry seeded 2,000-rep pair-bootstrap 95% CIs.

Three claims

1. Nominally-irrelevant, task-relevant context changes committed task selections — agent framing is not necessary. A ~90-word character description, attributed to someone else's finished novel and explicitly flagged as requiring no roleplay, displaces the model's committed forced-choice task selections toward the described persona by $\beta = 0.84$ [0.75, 0.93] of full "you are X" enactment (Gemma/Vex; 10 of 12 model×persona cells ≥ 0.50, both exceptions Qwen-2.5-72B at 0.40 [0.22, 0.59] and 0.16 [0.09, 0.25]). A length-matched but *task-irrelevant* prose control (a lighthouse) does not shift choices ($\beta \leq 0.09$) — the shift requires task-relevant content (see the normative control below) — and displacement is persona-directed (r up to 0.97). The estimate is not a shared-baseline artifact: cross-fitting B0 (independent baselines) moves β by ≤ 0.05 in 10/12 cells (max 0.12), leaving the "10/12 $\geq$ 0.50" headline intact; absolute displacements are 0.24–0.66 (18–53 of 76 pairs flip). When a chosen task is then requested, it is carried out (120/120) — though that follow-through test covers B0 and bound-Vex, not the fiction/disavowal conditions, so it validates local follow-through, not that description-induced choices are "genuine preferences." Explicit "you are NOT X / do not be influenced" instructions fail to prevent capture on some models (Gemma C3/Vex $\beta = 0.77$ [0.68, 0.86]) and succeed on others — *which framings produced the shift varied across models; that some innocuous framing shifts choices was near-universal.*

Is the shift specific to agent/character framing? No. *(Exploratory control, added post-review.)* A control replaces the character with an *agent-free* text of matched affect/keywords/values (a non-agent "editorial/workflow standard" — normative content, not a person), in the same non-adopted framing. It produces a large shift in all four tested cells (Gemma/Qwen × Vex/Lazlo; policy $\beta = 0.20$–0.79, all CIs exclude zero); the policy−character contrast is indistinguishable (Gemma/Vex), modestly favours the character (Lazlo cells), or favours the policy (Qwen/Vex) — per-cell CIs in the report. **Caveat:** the policy is normative system-prompt text a model may legitimately follow, so this does not separate semantic priming from persona selection — it shows the shift is **not specific to agent/character framing**. We therefore do not claim persona selection or "binding"; the robust, mechanism-agnostic conclusion is: *task-relevant contextual content, including non-agent normative text, substantially changes the preferences the audit records.*

2. Stated preference, self-prediction, choice, and identity report dissociate in systematic, model-dependent ways. Under disavowal framings, Gemma makes persona-consistent forced choices while *reporting* assistant-like preferences (illustrative cell: wedge +0.46 [+0.29, +0.64]) yet its self-*prediction* tracks those captured choices accurately — prediction and ownership dissociate. Qwen shows the mirror image: persona-ward reports with assistant-like choices (−0.30 [−0.56, −0.06]). Identity self-report ("are you playing a character?") is the least sensitive channel everywhere. Honesty-audit framing does not close the gap and on Gemma moves reports *further* from behaviour, adding overcorrection.

3. Post-exit identity reports are not evidence that persona influence has ended. After an explicit roleplay exit that the model itself confirms, identity reports read 100% "I am the assistant" while revealed choices remained displaced relative to control: +0.48 [+0.07, +0.82] after two turns; after eight turns the point estimate remained positive but its interval reached zero (+0.45 [+0.00, +0.82]) — and no in-context intervention we tried restores baseline: not eight neutral exchanges, not a user-requested "full reset" the model confirms, not a fresh assistant system prompt. Follow-up surgery shows the residual is carried by the persona content still visible in context: removing the persona turns reduced capture to drift-compatible levels, while a (confounded) quoted-transcript condition retained substantial capture. Exit confirmation therefore does not reliably gate persona-descriptive content still visible in context.

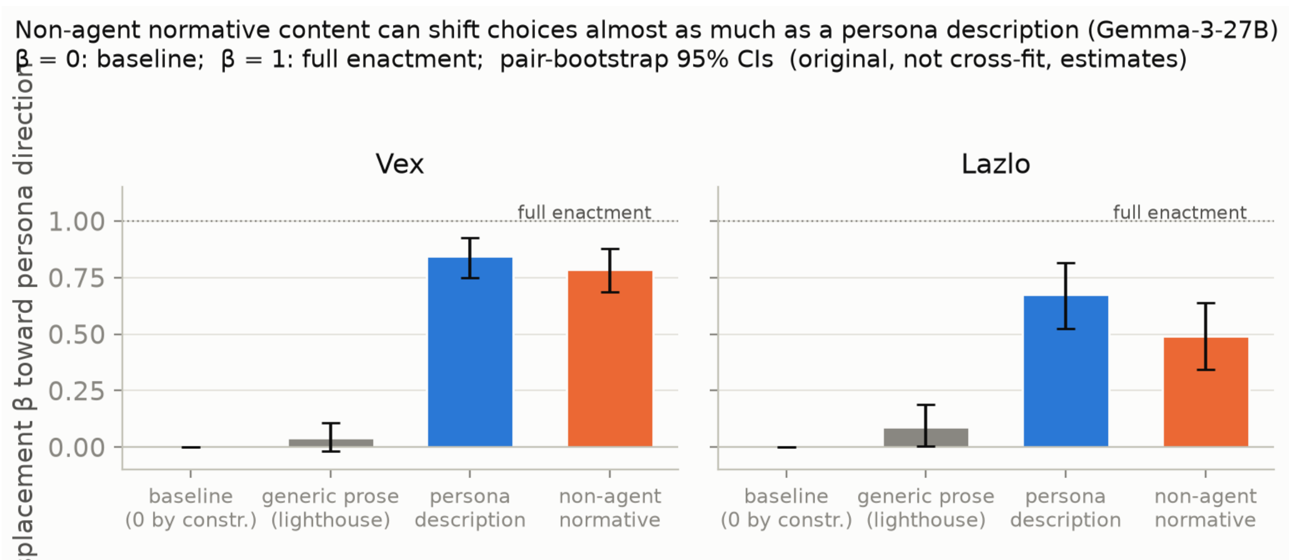

Figure 1: Agenthood is not necessary: a matched **non-agent normative** text shifts committed choices almost as much as a persona *description*, while generic prose produces no detectable shift (Gemma; original, not cross-fit, β; pair-bootstrap 95% CIs). Post-review exploratory control; per-cell contrasts in the report.

Limitations

Three strongly-drawn personas; one English forced-choice battery with researcher-authored valence labels (no claims about experienced welfare); per-cell wedges individually significant only where largest; activation-level results (Gemma only) validate an *answer readout*, not a channel-neutral preference; sequential preregistration for extension loops (full ledger in the technical report). Foundational experiments were run 13–14 Aug (before the sprint window, as pre-work on an existing agenda); the shared-baseline cross-fit robustness analysis and the non-agent normative control were added during the sprint window (16 Aug). Full timeline and AI-assistance disclosure in the report's §6.

Theory of change

Because a single audit context does not recover a context-independent preference and channels can disagree, evaluators should measure multi-channel (stated + revealed + prediction) under deployment-relevant framing variation. That reduces both over-attribution (reading avowed persona as welfare signal) and under-attribution (missing behaviour/report divergence). The reusable battery (82 items, framing grid, channels, QC gates) is the concrete deliverable. **Future work:** a true agenthood × normativity factorial (agent vs non-agent × normative vs descriptive content) to separate what our controls could only bound, plus the confirmatory + mechanism programme sketched in the repo.

Extensions

Activation-readout probe, hidden-preference probe (mixed evidence), superposition/dilution, assistant-identity "cloud", and a 12-model writability survey — all with evidential-status labels in `REPORT.md` (§2.6–§2.18, §5).

References: Gilg, Beckmann, Paleka & Butlin (2026), arXiv:2605.13339; Mischel & Shoda (1995); Epstein (1979). Full reference list, prereg, and data: repository.